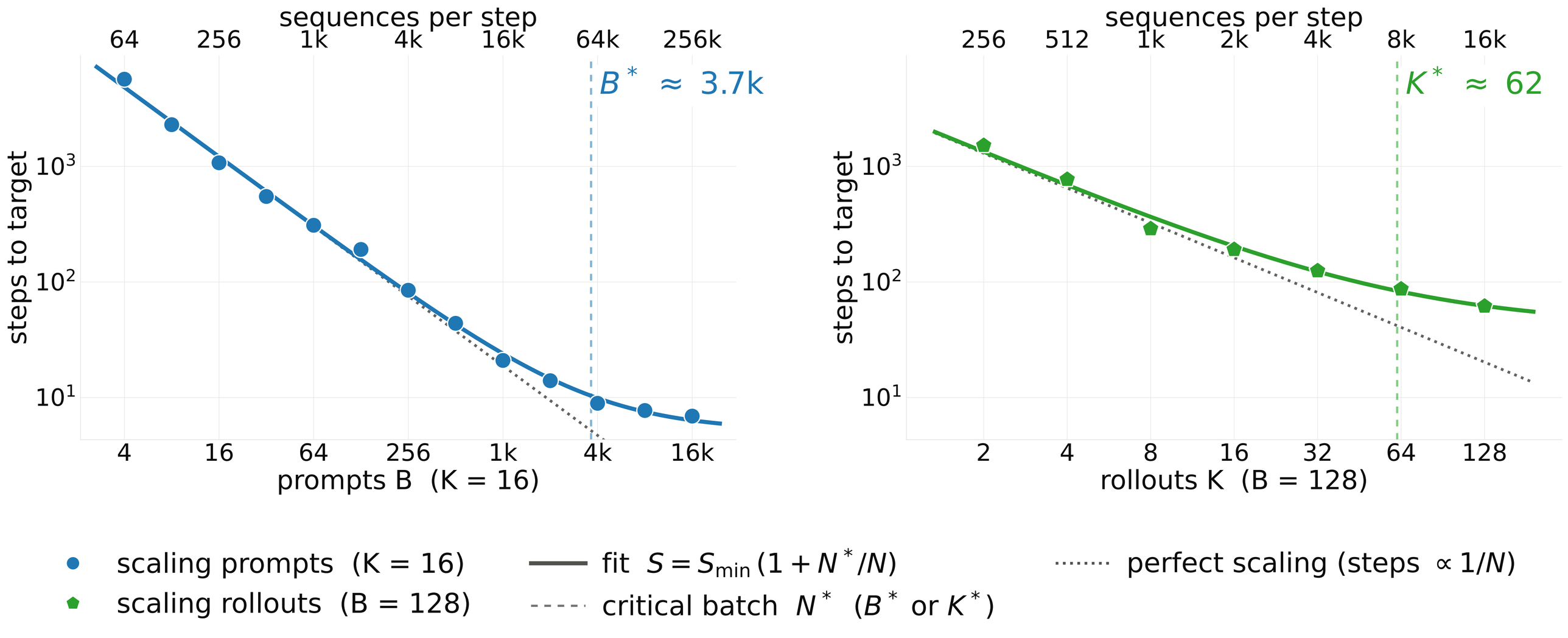
sequences per step
64
256
1k
4k
16k
64k
256k
$B^* \approx 3.7k$
steps to target
10^3
10^2
10^1
4
16
64
256
1k
4k
16k
prompts B (K = 16)
sequences per step
256
512
1k
2k
4k
8k
16k
$K^* \approx 62$
steps to target
10^3
10^2
10^1
2
4
8
16
32
64
128
rollouts K (B = 128)
scaling prompts (K = 16)
scaling rollouts (B = 128)
fit $S = S_{\min}(1 + N^*/N)$
critical batch N^* (B^* or K^*)
perfect scaling (steps $\propto 1/N$)